

Richard van Zyl — Skills Matrix (2026)

Professional years of experience, a SQL performance deep-dive, and a project-by-project breakdown per employer.

Years column. Professional years only, counted from paid work starting **April 2017** — nothing exceeds **9**. Formal study (Varsity College, 2014–2016) is not counted. Engagement-tied skills (PostgreSQL, Redis, MongoDB, Azure Functions) use that engagement's span, not the whole career.

Section	What's in it
Skills by area	Grouped tables, strongest first within each group
SQL performance deep-dive	Execution plans, indexing, locking, scale patterns
Employment & projects	Employer metadata, duties, and project cards

Skills by area

Within each table, rows are ordered by years (highest first).

Languages

Skill	Years
C# / .NET	9
T-SQL	9
PostgreSQL	7
TypeScript	5
JavaScript	5
KQL	5
Batch Script	5
Visual Basic .NET	3
Java	2

Data formats

Skill	Years
JSON	9
XML	8
XAML	7
YAML	6
HTML	3

Databases & data engineering

Skill	Years
SQL Server / SSMS	9
SQL performance tuning	8
Azure SQL	7
MySQL Workbench	6
Azure Data Studio	6
Redis (distributed caching)	5
SSRS	5
MongoDB	4
Table partitioning & archiving	4
In-Memory OLTP	4

Architecture & design

Skill	Years
Clean Architecture	9
Domain-Driven Design (DDD)	8
Microservices	7
Multi-tenancy / tenant isolation	7
CQRS	6
Event-driven architecture	5
Idempotency / exactly-once processing	4

Business analysis & solution design

Skill	Years
Business & problem-domain analysis	9
Requirements specification (BRS / URS / FRS / TRS)	9
Solution design & solution architecture	9
Technical documentation (audience-tailored)	9
Stakeholder engagement & translation	8
Process & system modelling (UML)	8

Backend frameworks & libraries

Skill	Years
Entity Framework	9
MediatR	7
AutoMapper	6
FluentValidation	6
ASP.NET	3
Circuit breaker (Polly)	3
SignalR / WebSockets	3

Frontend & UI

Skill	Years
Angular	6
Angular Material	6
Ajax	6
MVC	5
Node.js	5

Delivery practices

Skill	Years
Scrum	8
Unit testing (xUnit / NUnit / Moq)	6
OAuth2 / OpenID Connect / JWT	6
Integration testing (TestContainers)	5
Feature flags (Azure App Configuration)	4

Cloud, DevOps & tooling

Skill	Years
Visual Studio	9

Skill	Years
VS Code	9
GitHub	9
Azure DevOps (previously TFS)	9
Postman	9
Swagger	9
Enterprise Architect (UML)	8
Azure CI/CD	7
Docker	7
Pipeline-as-code (Azure DevOps YAML)	6
Message queues (Azure Service Bus / RabbitMQ)	6
Telerik	6
Azure Monitor / Log Analytics / App Insights	5
Microsoft Azure Insights	5
New Relic (APM, release tracking, alerting)	4
API Gateway (Azure API Management)	4
Elastic Stack (ELK) / Serilog	4
OpenTelemetry	3
Grafana	3
Graylog	3
DevExpress	3

Security & compliance

Skill	Years
Role-based access control (RBAC)	8
Data-protection-aware design (POPIA / GDPR)	5
SOX-aligned data segregation	4

Deep dive: SQL performance tuning

8 years · High-volume financial ledger systems integrating 150+ money processors across African markets.

Sub-skill	Focus
Execution plan analysis	SHOWPLAN, SET STATISTICS IO / TIME; spotting scans, key lookups, and high-cost operators
Indexing strategy	Clustered, non-clustered, filtered, and covering indexes; fragmentation and rebuild / reorganise decisions
Locking & concurrency	Isolation levels (READ COMMITTED → SNAPSHOT); deadlock detection via Profiler / Extended Events
Tuning patterns	Correlated subqueries → joins; INNER JOIN vs EXISTS; SARGable predicates; avoiding scalar UDFs
Design for scale	Date/tenant partitioning; indexed / materialised views; high-volume batch inserts (ledger entries)
In-Memory OLTP	Memory-optimised tables and natively compiled procedures on high-contention paths
Query Store	Regression tracking, forced parameterization, plan forcing
Health monitoring	sys.dm_os_wait_stats, sys.dm_db_index_usage_stats, sp_WhoIsActive

Employment & projects

Each block uses the same shape: metadata → duties → projects (role line, summary, tech). Side projects are labelled as such — they are not current employment.

Yuno Technologies (*side project — not employment*)

Field	Detail
Industry	Digital consultancy
Role	Stakeholder / Solutions Architect
Period	Apr 2026 – present
Operating	Partner-run; Richard contributes architecture, not day-to-day operations
Consulting	Yes (client names held private)

Duties - Stakeholder contributing architecture, backend engineering, and security design across client platforms. - **Current production:** four-plane deployment topology with one-way initiation; BFF holding session rather than tokens; response-only identity plane on Keycloak; containerised platform across environments. - **Phase 2 / planned:** inter-plane trust via mTLS / SPIFFE/SPIRE behind a documented interim. - **Under evaluation:** ledger-plane isolation via KVM-backed microVM separation so a container escape in an adjacent workload cannot reach the ledger. - ADRs with formal supersession across topology changes.

Tech: .NET · Vite / React · PostgreSQL · Redis · RabbitMQ · gRPC · Django / DRF · Azure Functions · Blob Storage · Keycloak · Prometheus / Grafana / ELK

Orchestration Platform (*side project — not employment*)

Field	Detail
Industry	Business orchestration
Role	Co-founder
Period	current
Operating	Side project; public at the same altitude as the site “Now” card

Duties - Deterministic business-orchestration platform on gRPC and standalone module assemblies; deterministic replay as the differentiator. - Owns the core engine, runtime, module execution, configuration and flow state, traces and audit logs.

Tech: gRPC · standalone module assemblies

Raging River Trading (Pty) Ltd

Field	Detail
Industry	Fintech / Gaming
Role	Solutions Architect / Technical Lead — Core Financial Systems (<i>payroll title: Software Developer</i>)
Period	Dec 2022 – Mar 2026
Consulting	No

Duties - Led the rewrite of a legacy banking platform into a white-label financial engine for deposits & withdrawals across African markets (including Betway and Jackpot City). - Engineered for horizontal and vertical scale under concurrent ledger load. - Integration platform for 150+ money processors — IronPython marshalling in .NET; rewrote per-integration isolation to stop shared thread-pool / socket exhaustion. - Migrated ~150 million records on a peak-trading Saturday with no New Relic Apdex drop.

Project — Core Financial Systems

Solutions Architect / Technical Lead · Team 1–4 · Backend

Full-scale rewrite to a white-label engine handling deposits & withdrawals for multiple brands. Horizontal scale with APIs that safely touch a concurrent ledger. Schema-per-brand under one SQL Server (separate databases priced out); transactional outbox to per-brand queues with dead-lettering.

Tech: Azure SQL · C# / .NET · Docker · RabbitMQ · EF Core

Project — Integration layer & back-office APIs

Solutions Architect / Technical Lead · Team 5 · Backend

Integration layer abstracting 150+ money processors. Back-office APIs for safe concurrent ledger access under high volume.

Tech: C# / .NET · Azure SQL

Project — High-availability & concurrency design

Solutions Architect / Technical Lead · Team 1 · Backend

HA architecture for concurrent financial transactions — idempotent attempt records with rowversion optimistic concurrency on the ledger.

Tech: C# / .NET · Azure SQL

MeterMo

Field	Detail
Industry	Utilities — automated electricity, water and gas metering & reporting
Role	Software Developer
Period	Apr 2022 – Dec 2022
Consulting	No
Leaving	Mandate complete; sought a larger engineering team

Duties

Platform modernisation across the metering estate. Migrated six projects from TFS to Azure DevOps and upgraded frameworks/packages; maintained two APIs, two cross-platform field apps (Xamarin, Cordova), and two websites.

Tech: C# / .NET · ASP.NET · Angular · Xamarin · Cordova · Azure DevOps · TFS

Dotcom Software Solutions

Field	Detail
Industry	Consulting (Fintech)
Role	Software Developer (Specialist Problem-Solver)
Period	Jan 2022 – Apr 2022
Consulting	Yes
Leaving	No longer remote / not a culture fit

Duties

Specialist problem-solver for challenges others could not resolve — full-stack fintech delivery for Nedbank, Standard Bank, and PSG Wealth. Business analysis including Schemes Bot monetisation design and payment validations.

Project — Schemes Bot (Standard Bank)

Software Developer · Team 5 · Backend · Jan–Feb 2022

BotFramework backend upgrade **to .NET 6 from .NET Standard**, including dependency updates and breaking-change fixes. Prior attempts had been rolled back; completed and live in production.

Tech: Azure SQL · BotFramework · C# / .NET 6 · MediatR · QnA Maker (*historical / retired*)

Project — PSG Wealth website

Software Developer / Analyst / Architect · Team 6 · Backend · Feb–Mar 2022

Angular Universal SSR for SEO — unblocked a two-year struggle in one week.

Tech: Docker · Angular · Bootstrap · Material · Node.js · C# / .NET 6 · Umbraco CMS

Payteq

Field	Detail
Industry	Fintech
Role	Senior Developer / Business Analyst / Team Lead
Period	Apr 2020 – Jan 2022
Consulting	No
Leaving	Career growth

Duties

Full-stack work on EDI, GoTrips, and Veriseal; business analysis with the BA team; team lead on the GoTrips rewrite, optimisation, and architecture.

Project — GoTrips

Software Developer / Analyst / Team Lead · Team 5 · Full stack · Jan 2021 – Jan 2022

Ground-up rewrite eliminating accumulated technical debt; coding standards, design principles, performance work, and automation for a maintainable system.

Tech: C# / .NET Core API · PostgreSQL · Azure CI/CD · FluentValidation · Angular · Azure Functions

Project — Veriseal

Software Developer / Analyst / Architect · Team 1 · Full stack · side project, migrated opportunistically

Independently migrated a critical compliance platform (KYC, AML, mass payments, bank-account verification) from legacy ASP.NET to .NET Core API with Angular 8.

Tech: C# / .NET Core · SQL · SSRS · Angular · ASP.NET · .NET Framework · FluentValidation · AutoMapper

Project — GoBills

Team Lead · Team 2 · Full stack · Apr 2020 – Jan 2021

Business-management solution (inventory, POS, financials) with sophisticated RBAC for permissions and client access.

Tech: C# / .NET Core API · Identity Framework · DI · custom CMS · Angular 8 · Bootstrap · Angular Material

Project — Interchange IDE & Insight Server

Development & maintenance · Team 2 · Support

Messaging-middleware integration platform with RBAC; ongoing SaaS support and maintenance.

Tech: SQL · C# / .NET Core · IP microservices / integrations

iPlan Global

Field	Detail
Industry	Supply chain / manufacturing / farming / industrial / automation
Role	Software Developer / Business Analyst
Period	Oct 2017 – Apr 2020
Consulting	Yes
Leaving	Career growth

Duties

Team and solo delivery; productivity automation; BA work across mining, industrial, supply chain, manufacturing, and sales — needs analysis, domain analysis, solution design and architecture.

Project — Consolidated Job Dashboard

Software Developer / Analyst / Architect · Team 1 · Full stack · May–Jun 2019

Supply-chain consolidation of job material requirements — replaced a manual, error-prone process.

Tech: VBScript · SQL

Project — Espresso Quote Application

Software Developer / Analyst / Architect · Team 1 · Full stack · Feb–Apr 2019

Multi-platform quoting for sales reps — on-the-spot quotes, email delivery, manager approval.

Tech: jQuery · SQL · C# / .NET · JavaScript · Entity Framework · MVC · Ajax

Project — ASP.NET Food Portal

Development & maintenance · Team 2 · Full stack · May–Sep 2019

Farmers log deliveries and poison-testing results with multi-location security and compliance.

Tech: ASP.NET · C# / .NET · XAML · MVVM

Project — Espresso RepCheckIn

Software Developer / Architect · Team 1 · Full stack · May–Sep 2019

Rep tracking and tiered priority visit lists for optimised field routes.

Tech: JavaScript · C# / .NET · jQuery · MVC · Ajax

Novigo (Pty) Ltd

Field	Detail
Industry	Recruitment
Role	Software Developer (C# / .NET)
Period	Apr 2017 – Oct 2017
Consulting	Yes (the company was the client)
Leaving	Contract ended

Duties

Business and problem-domain analysis; solution planning; framework design; system and server implementation; ERP maintenance; Linux server and domain/hosting support.

Project — Novigo Management System

Sole developer · Team 1 · Full stack · Apr–Sep 2017

Designed, built, and deployed a complete ERP from scratch, including server setup and IT support.

Tech: MySQL · C# / .NET · Linux · WinForms